

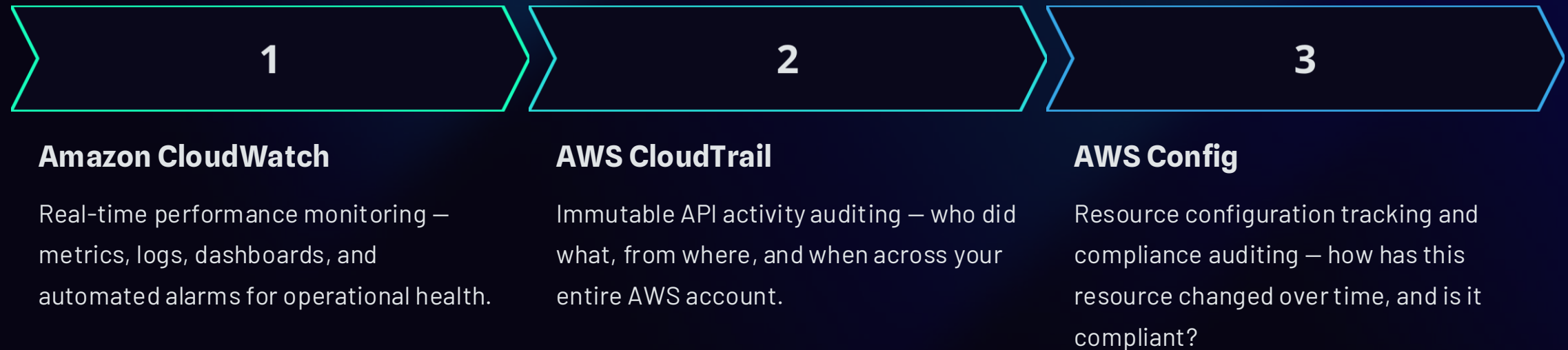
Continuous Monitoring & Auditing

CloudWatch · CloudTrail · AWS Config · ITNW 1009

Enterprise systems require two distinct perspectives running simultaneously: a **performance dashboard** showing if systems are healthy right now, and a **security ledger** showing who did what and when. Today, we master the three AWS tools that provide exactly these views — and learn to use them together for complete operational governance.



The Monitoring & Auditing Triad



❏ Monitoring tells you HOW systems run. Auditing tells you WHO changed them. You need both to manage a secure, reliable AWS environment professionally.

Key Vocabulary

Monitoring & Alerting

- **Monitoring:** Real-time collection and visualization of operational metrics and performance health markers across cloud resources.
- **Metrics:** Numerical data points — CPU utilization, disk I/O, network traffic — collected every 1–5 minutes.
- **CloudWatch Alarm:** An automated rule that monitors a metric and triggers action when it crosses a defined threshold.
- **Metric Filter:** A rule that extracts numerical metrics from plain text log data for graphing or alarm evaluation.

Auditing & Compliance

- **Auditing:** Recording and inspecting historical API activity and user actions for security, governance, and compliance.
- **API Event:** A recorded action triggered when any user, service, or script interacts with an AWS service via Console, CLI, or SDK.
- **Amazon CloudTrail:** Records every API call across your AWS account as an immutable JSON log entry.
- **AWS Config:** Continuously monitors and evaluates the configuration state and compliance of every AWS resource over time.



Amazon CloudWatch: System Observability

Is My System Healthy Right Now?

CloudWatch continuously gathers metrics from EC2, S3, RDS, and dozens of other services. If your web server hits 95% CPU utilization, CloudWatch sees it instantly. But it doesn't just display graphs — **it takes action**. You can configure a CloudWatch Alarm that says: "If CPU stays above 60% for more than 3 minutes, automatically launch two additional servers." That is how cloud systems heal themselves without human intervention.

CloudWatch — Four Core Capabilities



Metrics

Standard performance data automatically collected every 1–5 minutes – CPU load, disk I/O, network traffic, request count. Enable Detailed Monitoring for 1-minute intervals on critical resources.



Logs

Application and system log files gathered centrally in CloudWatch Log Groups. Organize logs by source – for example, `/aws/ec2/webserver`s for all web server text output.



Dashboards

Customizable real-time visual consoles displaying all your key metrics and alarms in one place. Build a shared dashboard for your operations team's NOC (Network Operations Center) wall screen.



Alarms

Automated rules that monitor a single metric over a time window and trigger actions – SNS notifications to on-call teams, EC2 Auto Scaling launches, or even Lambda function executions – the moment thresholds are crossed.

CloudWatch Logs & Metric Filters

Turning Text Into Actionable Numbers



System metrics tell you resource levels. Text logs tell you operational details — why something is happening. Using a **Metric Filter**, you set up pattern matching inside log streams. Every time the text `500 Internal Server Error` appears in your web server log, CloudWatch increments a custom counter. If that counter reaches 50 errors in one minute, CloudWatch triggers an alarm — **alerting your on-call team before users even notice the outage**.

AWS CloudTrail: The Immutable Flight Recorder

Who Did What, From Where, and When?

Every Action Is an API Call

Whether you click a button in the AWS Console, run an AWS CLI command, or execute an SDK script — **every action generates an API call**. CloudTrail captures each one as an immutable JSON log record and stores it in an Amazon S3 bucket.

CloudTrail logs contain:

- IAM Principal (who acted)
- Timestamp (when it happened)
- Source IP address (from where)
- Event Name (what was done)
- Response Elements (success or failure)

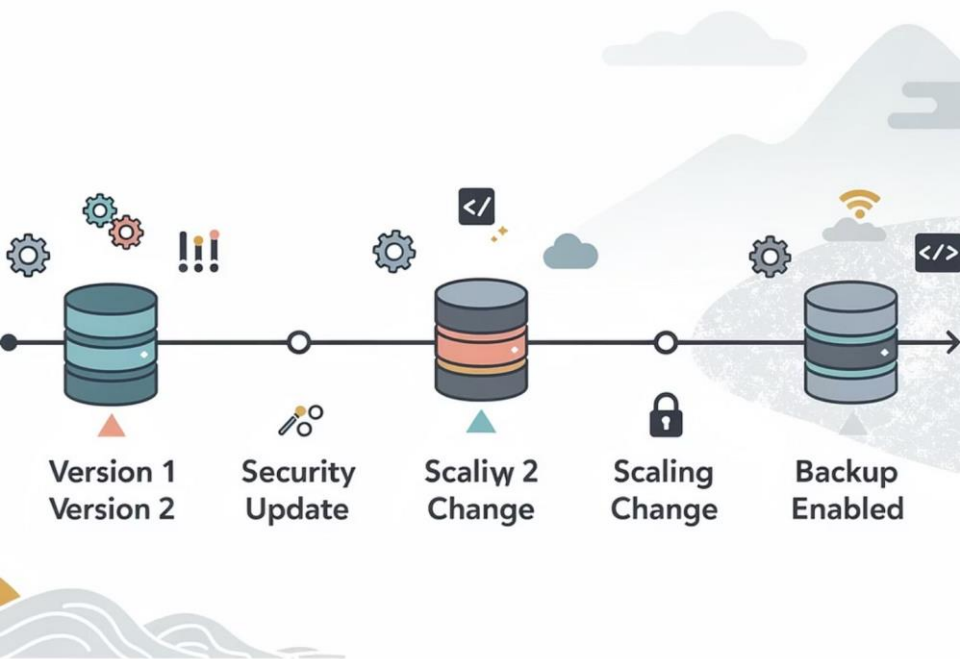
Why Immutability Matters

If an S3 bucket is deleted at 3:15 AM, CloudTrail tells you: *"IAM user 'DevJohn' logged in from IP 192.0.2.45, executed 'DeleteBucket', and succeeded."*

CloudTrail is essential for:

- Forensic security investigations
- Regulatory compliance — HIPAA, PCI-DSS, SOC 2
- Insider threat detection
- Incident root-cause analysis

Standard users cannot alter CloudTrail logs — ensuring complete accountability across your entire organization.



AWS Config: Resource Compliance Tracking

How Has This Resource Changed — and Is It Compliant?

While CloudTrail tells you *who* clicked the button, **AWS Config records what the resource looked like before and after that click** — building a full historical timeline of every configuration state. You define **Config Rules** as baseline standards. If a developer accidentally opens SSH port 22 to the entire Internet (0.0.0.0/0), AWS Config detects the violation immediately, flags the resource as Non-Compliant, and can trigger an automated remediation script to close the port — all before a human even notices.

The Three-Tool Comparison

Service	Primary Question	Core Data Type	Primary Use Case
Amazon CloudWatch	"Is the system healthy <i>right now</i> ?"	Performance Metrics & Log Files	Autoscaling, Alerting, Operational Dashboards
AWS CloudTrail	"Who made the API call?"	API Activity Event Logs (JSON)	Security Auditing, Incident Forensics, Governance
AWS Config	"Is resource state <i>compliant</i> ?"	Configuration Timelines & Rules	Compliance Auditing, Asset Inventory, Auto-Remediation

📌 Exam tip: Performance/CPU/logs → CloudWatch. API calls/who deleted something → CloudTrail. Configuration changes over time/compliance rules → AWS Config. These three rarely overlap.

The High-Security Bank Analogy

Making the Triad Unforgettable



CloudWatch = Building Sensors

Environmental sensors inside the bank vault — measuring temperature, air pressure, power consumption, and teller queue length. If vault temperature rises above 85°F, an automated alarm sounds to alert the facilities team immediately. Continuous real-time health data.



CloudTrail = Security Camera Logs

Time-stamped video recording and front-desk visitor ledger. Records every individual who enters, every keycard swipe at a secure door, every safe combination entered, and every transaction signed by a manager. Immutable evidence for investigations.

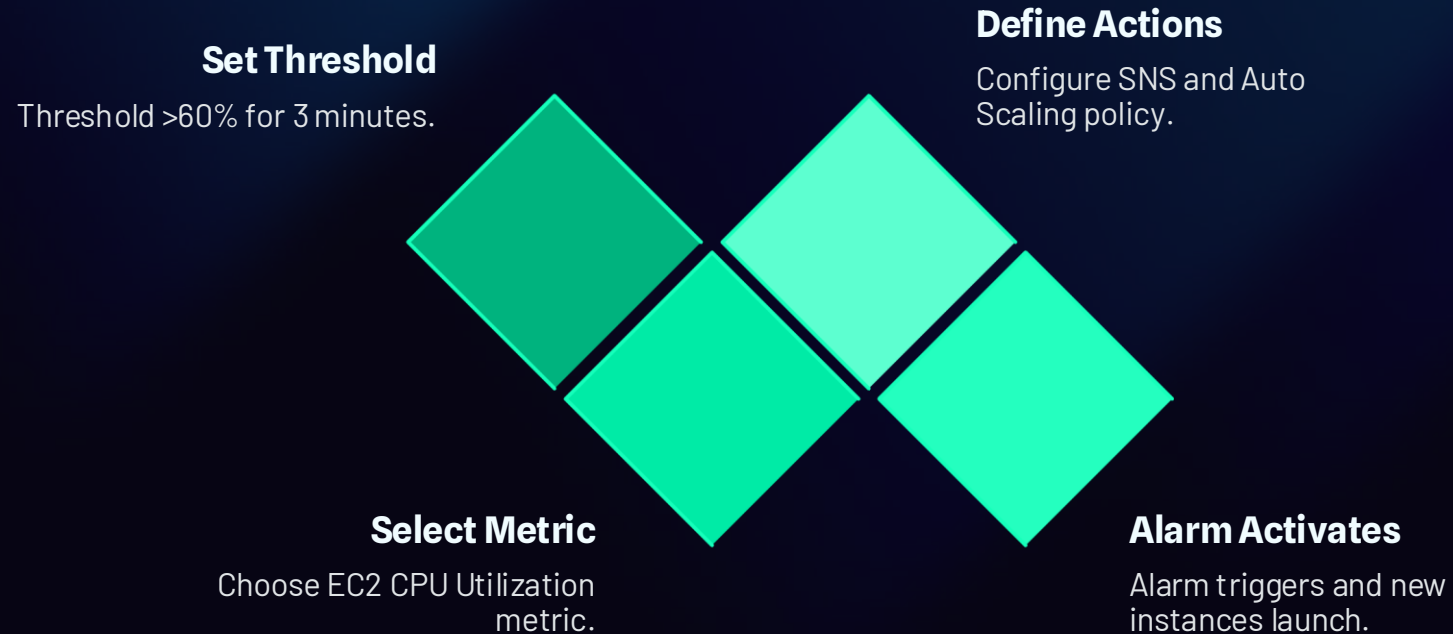


AWS Config = Structural Blueprint

The master architecture blueprint and inventory registry. Tracks the physical structural state over time — recording when a lock was upgraded from a 3-bolt to a 5-bolt system, verifying all safety deposit boxes meet state security standards. Compliance over time.

Configuring a CloudWatch Alarm

Hands-On Skill: CPU Threshold → Auto Scaling



This is a **core hands-on skill for Cloud Support Associates and Solutions Architect Associates**. Configuring CloudWatch Alarms tied to Auto Scaling policies is how production systems at every major AWS customer maintain performance during traffic spikes – automatically, without a human waking up at 3 AM to scale the fleet manually.

CloudTrail & Config in Practice

Forensic Investigation Scenario

The Incident

At 2:47 AM on a Tuesday, an S3 bucket containing sensitive customer data is modified to allow public access. The Security team receives an automated Config alert. An investigation begins immediately.

The Investigation

1. **AWS Config** shows the bucket's public access setting changed from `Blocked` to `Enabled` at exactly 02:47:12 AM.
2. **AWS CloudTrail** identifies the API call: `PutBucketAcl` executed by IAM user `intern-alex` from IP address `203.0.113.88`.
3. Security immediately revokes `intern-alex`'s credentials, re-blocks public access, and files an incident report — all with a complete, irrefutable evidence trail.

Without CloudTrail, you'd know the bucket was public. **With CloudTrail, you know exactly who, when, and from where.**

Certification Exam Strategy

Monitoring & Auditing — Exam Tips



CloudWatch = Health & Performance

Any exam question mentioning CPU spikes, memory utilization, application logs, auto-scaling triggers, dashboards, metric alarms, or operational observability → **Amazon CloudWatch** is the answer.



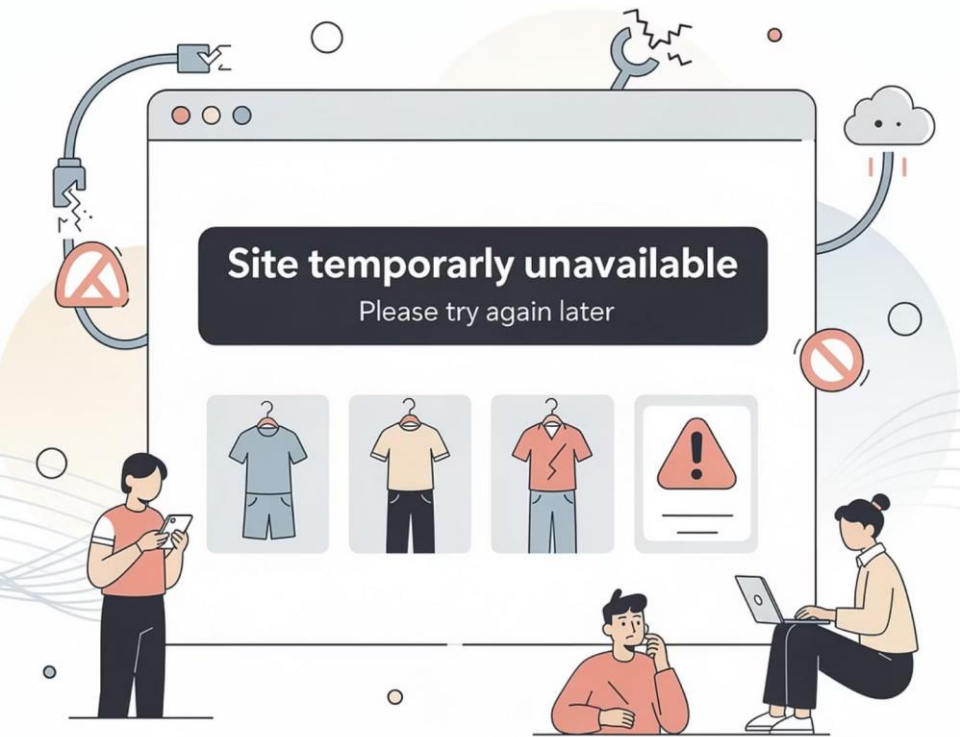
CloudTrail = Who & When

Any question about tracking user activity, recording API calls, determining who deleted a resource, compliance logging for HIPAA/PCI-DSS, or forensic investigation of an incident → **AWS CloudTrail** is the answer.



AWS Config = Compliance & Change History

Any question about tracking how a resource's configuration changed over time, evaluating compliance against security rules, automated remediation of non-compliant resources, or configuration drift detection → **AWS Config** is the answer.



The MegaCart Case Study

What Happens When You Ignore Trusted Advisor

MegaCart Retail — 50,000+ product SKUs — suffered a **catastrophic Cyber Monday collapse**. At 12:00 PM, traffic spiked to ten times normal capacity. The monolithic app on a single oversized EC2 instance hit 100% CPU. Session data written to local server RAM was wiped when the instance was restarted. The 5-hour blackout cost an estimated **\$420,000 in unfulfilled purchases**. Post-incident review found the team had been *ignoring* Trusted Advisor warning emails for weeks — database storage over 85% full, no read replicas, single-AZ deployment flagged as red. The alarms were there. Nobody listened.

MegaCart — Structural Failures Mapped to Pillars



Reliability Violated

Single EC2 instance, single-AZ database, no Auto Scaling, no Elastic Load Balancer. One failure point triggered total collapse.



Performance Efficiency Violated

Product images, CSS, and banners served from attached EBS volume — saturating disk I/O and starving database connection pools under load.



Operational Excellence Violated

No CI/CD pipeline, no deployment automation. Manual SSH changes to live production systems — a typo during peak hours is catastrophic.



Cost Optimization Violated

Disabled database backups "to save \$15/month." The resulting \$420,000 outage is the most expensive cost optimization decision imaginable.

Unit Summary

What You've Mastered

DNS & CDN

Route 53 resolves domain names to IPs through a 4-step resolver chain. Routing policies (Weighted, Latency, Failover, Geolocation) manage intelligent traffic distribution. CloudFront's Edge Locations eliminate distance latency through caching, OAC, and TTL management.

Well-Architected

The six pillars (Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability) provide a balanced framework for every architecture decision. HA vs. FT trade-offs, DR strategies (RPO/RT0), and Trusted Advisor's continuous auditing keep systems resilient and cost-conscious.

Monitoring & Auditing

CloudWatch answers "Is the system healthy?" with metrics, logs, dashboards, and alarms. CloudTrail answers "Who did what and when?" with immutable API event logs. AWS Config answers "Is resource state compliant?" with configuration timelines and automated remediation rules.

You now have the foundational knowledge to **design, defend, and monitor AWS architectures** like a Cloud Support Associate. Well done — keep going! 🎓